

Lecture - 14

Bayesian Estimation & Uncertainty

- Till, statistical inference is estimating unknown parameter θ , from data $X_1, \dots, X_n$. Let's say $X_1, \dots, X_n \sim f(x|\theta)$. After "observed", $X_1 = x_1, \dots, X_n = x_n$. We want to learn θ .

Frequentist

θ is unknown, fixed

$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta; x)$
then quantify uncertainty
using $SE(\hat{\theta})$ and CI

Bayesian

treat uncertainty of θ
using bayesian distribution
 $\pi(\theta|x)$ inference gives
posterior distribution

- Bayesian inference req. probab. models for 2 things we specify, $\theta \sim \underbrace{\pi(\theta)}_{\text{prior distribution}}$ and $X|\theta \sim \underbrace{f(x|\theta)}_{\text{data model}}$.

$L(\theta; x) = f(x|\theta) =$ likelihood once x is obs.

- The prior is $\pi(\theta) \rightarrow$ what we assume about θ before observing the current data.
- Advantages of bayesian inference
 - $\rightarrow$ it incorporates prior information
 - $\rightarrow$ prior knowledge + observed data
- disadvantage $\rightarrow$ results are sometimes sensitive to the choice of prior.

- Bayes Theorem $\rightarrow \pi(\theta|x) = \frac{f(x|\theta) \pi(\theta)}{\int f(x|t) \pi(t) dt}$

does not depend on θ .

$$\pi(\theta|x) \propto L(\theta; x) \pi(\theta)$$

posterior $\propto$ likelihood $\times$ prior

updated knowledge = information from data $\times$ prior information

- $L(\theta; x) = f(x|\theta) \rightarrow$ what values of θ makes the observed data more plausible.

$\pi(\theta|x)$ is probability distribution, it says posterior after observing the data what is our uncertainty regarding θ .

Posterior = $\frac{\text{likelihood} \times \text{prior}}{\text{normalizing constant}}$

- $X_1, \dots, X_n \sim N(\mu, \sigma^2) \rightarrow$ known $\rightarrow$ estimate

$\mu \sim N(\mu_0, \tau^2) \rightarrow$ prior variance
prior mean

derivation

$(\mu|x) \sim N(\mu_n, \tau_n^2)$

$$\frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2}$$

$$\mu_n = \frac{\frac{\mu_0}{\tau^2} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}$$

- $L(\mu; x) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right]$

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$$

does not depend on μ .

$$L(\mu; x) \propto \exp \left(-\frac{n}{2\sigma^2} (\mu - \bar{x})^2 \right)$$

prior $\pi(\mu) \propto \exp \left(-\frac{1}{2\tau^2} (\mu - \mu_0)^2 \right)$

$$\therefore \pi(\mu|x) \propto L(\mu; x) \times \pi(\mu)$$

so, $(\mu|x) \sim N(\mu_n, \tau_n^2)$

- precision = $\frac{1}{\text{variance}}$

$$\text{posterior precision} \geq \frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2}$$

posterior precision = prior precision + data prec.

$$n \uparrow \Rightarrow \tau_n^2 \downarrow$$

more info $\rightarrow$ larger precision $\rightarrow$ smaller variance

$$\mu_n = \omega \mu_0 + (1 - \omega) \bar{x} \quad \text{where} \quad \omega = \frac{\sigma^2}{\sigma^2 + n\tau^2}$$

so, posterior mean sits btw. μ_0 and $\bar{x}$.

$\therefore$, bayesian learning,

prior belief $\rightarrow$ updated belief after data obs.

if $n \rightarrow \infty$, $\omega \rightarrow 0$, $\mu_n \rightarrow \bar{x}$.

as data increases, data dominates the prior.

if τ^2 is small, ω becomes large, μ_n is close to μ_0 .

eg. $X_i \sim N(\mu, 4)$, $\sigma^2 = 4$
 $\mu \sim N(10, 9)$, $\mu_0 = 10$, $\tau^2 = 9$, $n = 5$
 $\bar{x} = 14$

$$\frac{1}{\tau_n^2} = \frac{1}{9} + \frac{5}{4} = \frac{4 + 45}{36} = \frac{49}{36} \rightarrow \tau_n^2 = \frac{36}{49} = \frac{6}{7}$$

$$\tau_n^2 = 0.735, \quad \tau_n = 0.857$$

$$\omega = \frac{\sigma^2}{\sigma^2 + n\tau^2} = \frac{4}{4 + 5(9)} = \frac{4}{49}$$

$$\mu_n = \frac{4}{49} \times 10 + \frac{45}{49} \times 14 = \frac{40}{49} + \frac{630}{49} = \frac{670}{49}$$

$$\mu_n = 13.673$$

$$\mu|x \sim N(13.673, 0.735)$$

$$\begin{array}{|c|} \hline 10 < 13.673 < 14 \\ \hline \mu_0 < \mu_n < \bar{x} \\ \hline \end{array} \rightarrow \text{takeaway}$$

- once we have $\Pi(\theta|x)$

$$\hat{\theta}_{PM} = E(\theta|x)$$

expected value of the posterior / posterior mean

$$\text{if } (\theta|x) \sim N(\mu_n, \tau_n^2)$$

$$\text{then, } \hat{\theta}_{PM} = \mu_n$$

$$\hat{\theta}_{med} = \text{median}(\theta|x)$$

$$\text{if the value } m \Rightarrow P(\theta \leq m|x) = 0.5$$

for sym. distributions like normal, mean = median
 but for skewed, it might differ

- MAP (Maximum A Posteriori)

$$\hat{\theta}_{MAP} = \arg \max_{\theta} \Pi(\theta|x) \rightarrow \text{finds the value of } \theta \text{ where posterior density is largest}$$

$$\pi(\theta|x) \propto L(\theta; x) \times \pi(\theta)$$

$$\hat{\theta}_{MAP} = \arg \max_{\theta} L(\theta; x) \pi(\theta)$$

$$\text{or } \hat{\theta}_{MAP} = \arg \max_{\theta} [L(\theta; x) + \log \pi(\theta)]$$

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta; x)$$

$$\hat{\theta}_{MAP} = \arg \max_{\theta} L(\theta; x) \pi(\theta)$$

$$MAP = MLE \times \text{prior}$$

if prior is flat $\Rightarrow \pi(\theta) \propto 1$
then $\hat{\theta}_{MAP} = \hat{\theta}_{MLE}$

- Whether a distribution is flat depends on the parameterization. $\theta' = g(\theta)$ is a non-linear transformation. A prior that is flat in θ may not be flat in θ' . MAP estimates are not invariant under non-linear transformations.

$$\text{say } \phi = g(\theta), \text{ then } p_{\phi}(\phi) = p_{\theta}(\theta) \left| \frac{d\theta}{d\phi} \right|$$

The jacobian term can alter which pt. has the highest density. $g(\hat{\theta}_{MAP}) \neq \hat{\phi}_{MAP}$. This is because MAP maximizes the density, but density changes under coordinate transformation.

$$\text{Let's say } \theta \sim N(0, \sigma^2), \pi(\theta) \propto \exp\left(-\frac{\theta^2}{2\sigma^2}\right)$$

$$\text{so, } \log \pi(\theta) = -\frac{\theta^2}{2\sigma^2} + C$$

$$\text{recall, } \hat{\theta}_{MAP} = \arg \max_{\theta} [L(\theta; x) + \log \pi(\theta)]$$

$$\left| \hat{\theta}_{MAP} = \arg \min_{\theta} [-L(\theta; x) + \lambda \theta^2] \right|, \lambda = \frac{1}{2\sigma^2}$$

so, gaussian prior $\Leftrightarrow$ quadratic regularization

$$\text{Ridge } \min_{\beta} \left[\text{RSS} + \lambda \sum_j \beta_j^2 \right] \quad \beta_j \sim N(0, \tau^2)$$

= MAP estimator

- with $\hat{\theta}$, we have $\pi(\theta|x)$
 where $\hat{\theta}$ is a most likely $\rightarrow$ MAP
 what is the posterior center $\rightarrow$ posterior mean/mode
 how uncertain we are $\rightarrow$ posterior variance
 which value of θ is plausible $\rightarrow$ credible interval.

- A $100(1-\alpha)\%$ is credible interval $[L, U]$ s.t.

$$P(L \leq \theta \leq U | x) = 1 - \alpha$$

for 95% credible interval $P(L \leq \theta \leq U | x) = 0.95$
 let's say $(\theta | x) \sim N(\mu_n, \tau_n^2)$
 95% $\Rightarrow \mu_n \pm 1.96 \tau_n$
 $\rightarrow$ posterior S.D.

eg. $(\mu | x) \sim N(13.674, 0.735)$, $\tau_n = 0.857$
 95% credible interval
 $[11.994, 15.354]$

$$P(11.994 \leq \mu \leq 15.354 | x) = 0.95$$

eg. $[2.1, 3.4] \Rightarrow P(2.1 \leq \theta \leq 3.4 | x) = 0.95$
 given the obs. data & prior, there is a 95% posterior probab. that θ lies btw. 2.1 & 3.4.
 but a frequentist can't say $P(2.1 \leq \theta \leq 3.4) = 0.95$
 because for them θ is fixed! They can say that
 if we collect samples repeatedly and build the
 interval using same procedure, approx. 95% of
 those would have true value of θ .

credible interval $\Rightarrow$ probab. abt. the param.
 confidence interval $\Rightarrow$ probab. abt. the procedure

frequentist $\Rightarrow L(\theta) \Rightarrow \hat{\theta}_{MLE} \Rightarrow p(\hat{\theta} | \theta) \Rightarrow CI$
 bayesian $\Rightarrow L(\theta) \star \pi(\theta) \Rightarrow \hat{\theta}_{MAP}$, etc.